

2024531115

Digital Circuits Homework

Spring 2026

Instructor: Chenxi Xiao

TA: Ruilin Zhang

zhangrl2023@shanghaitech.edu.cn

Due: 23:59, March 27, 2026 (W4 Friday)

Instructions

- Solutions should be hand-written. Show necessary steps clearly.
- Unless otherwise specified, all binary calculations should be carried out directly in binary.
- Upload to Blackboard as a single pdf file.
- Late submission may result in deduction in score.
- Total Score: 100 pts.

Problem 1. (14 pts) Convert the following numbers to the required base.

1. $(0110101101)_2 = (\quad 429 \quad)_{10}$

2. $(\underline{1001110011})_2 = (\quad 1163 \quad)_8$

3. $(\underline{1101011011})_2 = (\quad 35B \quad)_{16}$

4. $(B4E)_{16} = (\quad 2894 \quad)_{10}$

5. $(607)_8 = (\quad 110000111 \quad)_2$

6. $(117)_{10} = (\quad 1110101 \quad)_2$

7. $(254)_8 = (\quad 172 \quad)_{10}$

Problem 2. (12 pts) Perform the following calculations using **8-bit two's-complement representation**. All numbers in the given expressions are in decimal.

For each part:

- convert all operands to 8-bit binary,
- perform the calculation step by step,
- give the final 8-bit result,
- and state whether overflow occurs.

(1) $35 - 19$

$$\begin{aligned} &= 00100011 - 00010011 \\ &= 00100011 + 11101101 \\ &= 00010000 \end{aligned} \quad \text{no overflow}$$

(2) $27 - 58$

$$\begin{aligned} &= 00011011 - 00111010 \\ &= 00011011 + 11000110 \\ &= 11100001 \end{aligned} \quad \text{no overflow}$$

Problem 3. (10 pts)

1. Compute

$$(10110100)_2 \oplus (01101110)_2$$

Show the bitwise XOR process clearly.

$$\begin{array}{r} 10110100 \\ 01101110 \\ \hline 11011010 \end{array}$$

2. Compute

$$(001101)_2 \times (000111)_2$$

Show the multiplication procedure entirely in binary.

$$\begin{array}{r} 001101 \\ \times 000111 \\ \hline 001101 \\ 0011010 \\ 00110100 \\ 00000000 \\ 000000000 \\ 0000000000 \\ \hline 000001011011 \end{array}$$

Problem 4. (16 pts) Complete the following truth table for the Boolean expressions:

$$P_0 = A'B + AB'$$

$$P_1 = A'B'C + ABC' + BCD$$

$$P_2 = A' + B' + C' + D' = \overline{ABCD}$$

$$P_3 = AB'CD'$$

A	B	C	D	P_0	P_1	P_2	P_3
0	0	0	0	0	0	1	0
0	0	0	1	0	0	1	0
0	0	1	0	0	1	1	0
0	0	1	1	0	1	1	0
0	1	0	0	1	0	1	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	1	0
0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	0	1	0
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0
1	1	0	0	0	1	1	0
1	1	0	1	0	1	1	0
1	1	1	0	0	0	1	0
1	1	1	1	0	1	0	0

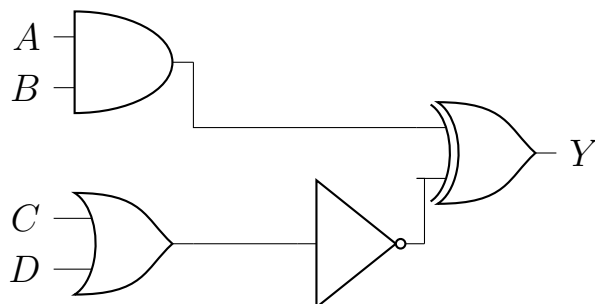
Problem 5. (10 pts) Simplify the following Boolean expression as much as possible using Boolean algebra. Show your derivation clearly.

$$F = XYZ' + X'YZ' + XYZ + XY'Z$$

$$= (x + \bar{x}) Y \bar{z} + (\bar{y} + y) x z$$

$$= Y \bar{z} + x z$$

Problem 6. (14 pts) Determine the Boolean expression for the output Y as a function of inputs A , B , C , and D for the following logic circuit, and write down its truth table. You do **not** need to simplify the final expression.



$$Y = (AB) \oplus \overline{(C+D)} = (AB) \oplus (\bar{C}\bar{D})$$

Y	A	B	C	D
1	0	0	0	0
0	0	0	1	0
0	0	0	0	1
0	0	0	1	1
1	0	1	0	0
0	0	1	1	0
0	0	1	0	1
0	0	1	1	1

Y	A	B	C	D
1	1	0	0	0
0	1	0	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
1	1	1	1	0
1	1	1	0	1
1	1	1	1	1

Problem 7. (14 pts) **Universal-gate implementation.**

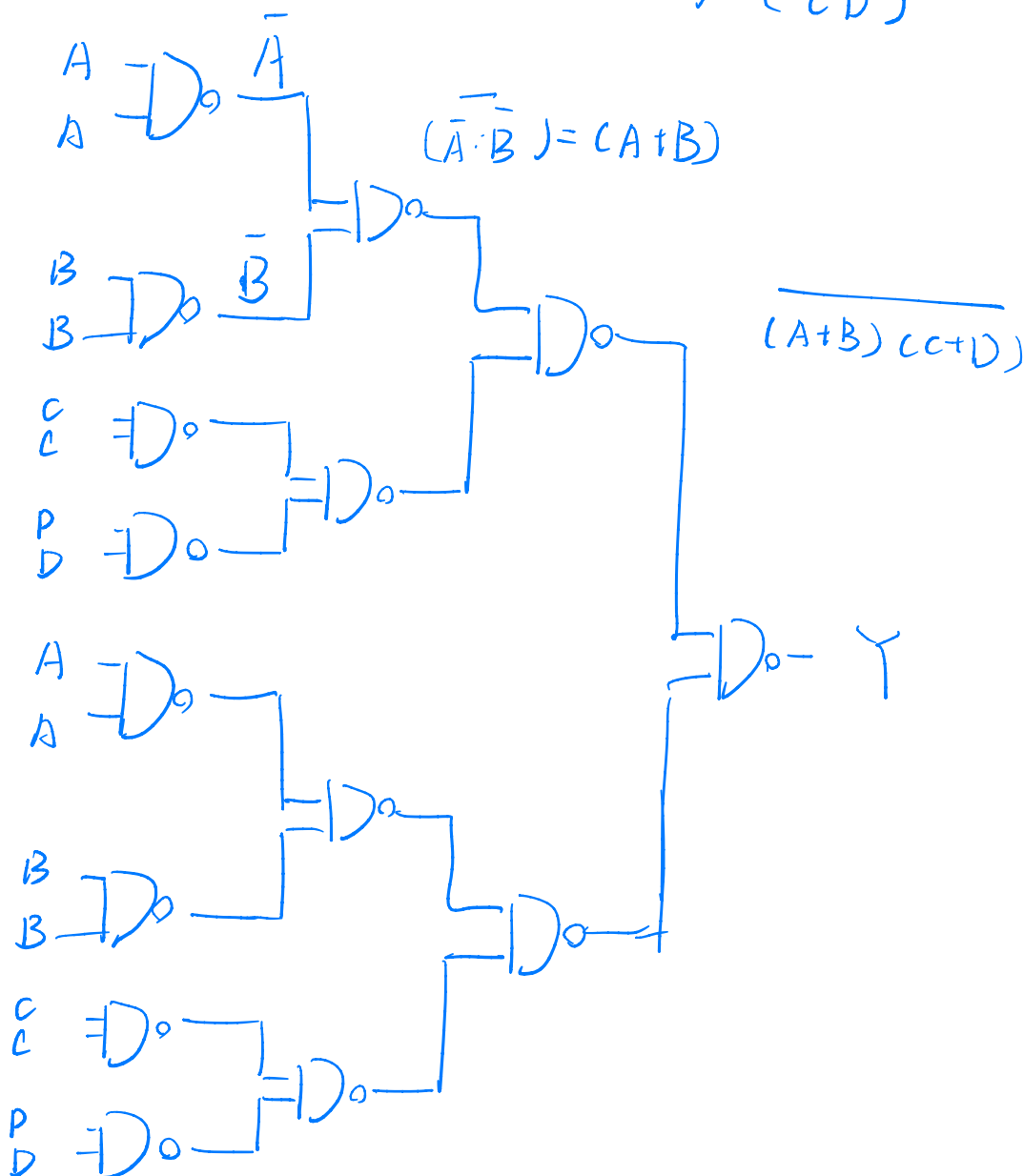
Implement the Boolean function

$$Y = (A + B)(C + D)$$

using **only 2-input NAND gates**.

- First derive an equivalent NAND-only implementation step by step.
- Then draw the final logic circuit.
- Clearly label intermediate signals.
- Use as few gates as possible.

$$Y = (A + B)(C + D) = (\overline{\overline{A} \cdot \overline{B}}) \cdot (\overline{\overline{C} \cdot \overline{D}})$$



Problem 8. (10 pts)

A two-input logic device was tested, and the following steady-state dataset was obtained.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

In a timing test, the following propagation delays were observed:

- when the input event causes the output to change from 1 to 0, the delay is 12 ns;
- when the input event causes the output to change from 0 to 1, the delay is 9 ns.

1. Write the Boolean expression for Y .
2. State which delay corresponds to t_{PHL} and which corresponds to t_{PLH} .

1. $Y = \overline{A}B$

2. $t_{PHL} = 12 \text{ ns} \quad 1 \rightarrow 0$

$t_{PLH} = 9 \text{ ns} \quad 0 \rightarrow 1$